

ORIGINAL RESEARCH

Quantifying new threats to health and biomedical literature integrity from rapidly scaled publications and problematic research

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Abstract

Background and Objectives: The last 3 years have seen an explosion in published manuscripts analyzing open-access health datasets, in many cases presenting misleading or biologically implausible findings. There is a growing evidence base to suggest that this is due in part to artificial intelligence-assisted and formulaic workflows, and publishers are responding by discouraging submissions employing open-access health datasets.

Methods: Here we use a scientometric analysis to investigate which datasets have seen publication rates deviate from previous trends, especially where this coincides with changes to author geographical origins and increases in formulaic titles.

Results: Across 36 datasets, we identify nine showing hallmarks of paper mill exploitation (FDA Adverse Event Reporting System, National Health And Nutrition Examination Survey, UK Biobank, FinnGen, the Global Burden of Disease Study, Medical Information Mart for Intensive Care, China Health and Retirement Longitudinal Study, Centers for Disease Control and Prevention Wide-ranging Online Data for Epidemiologic Research, and TriNetX). These nine datasets had, in 2025, a combined publication count of 23,005 indexed in the OpenAlex database. This represents an excess of 11,577 publications above the AutoRegressive Integrated Moving Average forecast trend, and is a 3.0 $\times$ -fold change on the 7655 publication count for these nine datasets in 2022. We also identified a notable difference in the fold change for China (4.2 $\times$) vs. the rest of the world (1.9 $\times$) and an increase in formulaic titles.

Conclusion: These findings highlight potential risks to research integrity in areas such as public health and drug safety, and especially to the accessibility and interoperability principles central to Open Science and Findable, Accessible, Interoperable and Reusable data practices. We argue that permissive open-access data policies naturally facilitate exploitative workflows and that these findings add to the case for the safeguarding mechanisms to preserve the goals of Open Science. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Metascience; Paper mills; Artificial intelligence; FAIR guiding principles; Open science; Integrity

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What is new?

Key findings

- In this work we systematically review 36 major resources for sudden changes in trend rates of publications indexed in the OpenAlex database. We quantify the explosion of low-value publications exploiting health data—an estimated annual excess of 11,577 by 2025 compared with previous trends— and demonstrate that this has been accompanied by a dramatic shift in geographic origin and an increase in formulaic titles, including in sensitive fields such as public health and drug safety.

What this adds to what is known?

- Existing descriptions of mass-produced and artificial intelligence (AI)-assisted manuscripts have focused on individual problem areas, such as National Health and Nutrition Examination Survey or Mendelian randomization. Our goal is to systematically gather evidence of research mass produced by paper mills and illustrate that the problem can extend to any field with large open-access datasets, challenging the integrity of the scientific literature. Our code and methods are freely available and can be extended as frequently as necessary.

What is the implication and what should change now?

- Our findings show a growing problem with low-quality, AI-assisted and mass-produced manuscripts, which by definition are evading existing checks and balances in the scientific literature and are being accepted by indexed journals. This is a particular issue at a time when generative AI is making health information more widely available to the general public, especially when such tools are trained on peer-reviewed literature, a resource which should be high quality, credible, and self-correcting. The exploitation of data presents a challenge to the principles of Open Science and the Findable, Accessible, Interoperable and Reusable Guiding Principles, and if left unchecked over the longer term could lead to disincentives to data sharing and loss of credibility. In the short term, we believe that increased awareness of data exploitation will be essential to strengthen editorial decisions, peer review, and postpublication correction, as bulwarks of self-correcting science.

1. Introduction

Generative artificial intelligence (AI) (GenAI) and other automation tools have the potential to transform productivity in biomedical and health research. New technologies can, however, also be exploited in problematic ways, and paper mills (entities which mass-produce manuscripts for purchase) are particularly likely to benefit from such productivity gains as their business model relies on the large-scale authoring of inevitably low-quality or completely fabricated manuscripts [1–3]. These issues have been identified previously. Notably, the exploitation of the National Health And Nutrition Examination Survey (NHANES) data resource has been highlighted by targeted analysis of individual manuscripts [4,5], with problematic research practices including lack of false discovery correction, selective data usage/hypothesizing after the results are known (also known as HARKing), and data dredging to maximize manuscript counts irrespective of the plausibility of the findings [6]. Investigating individual publications is time-consuming and can be slow to react to new trends, and so trend analyses can also be helpful when identifying problematic research practices, especially when resources are scarce [7]. As well as the rapid growth in the number of publications citing NHANES [8], concerning trends have also been identified in works poorly employing two-sample Mendelian randomization applied to openly available Genome-Wide Association Study (GWAS) data [9,10]. To our knowledge, however, there has not yet been a systematic attempt to quantify the growth in exploitation across the wider field of open access datasets. This has the potential to be a particular problem for datasets in the field of health and epidemiology, especially those that are Findable, Accessible, Interoperable and Reusable (FAIR) compliant or with Application Programming Interface (API) supported access, which may be vulnerable to AI-supported and pipeline-driven mass production of papers.

These issues are, of course, not new. Genomics provides a case study of a field that experienced an explosion of results, often lacking in biological plausibility or which could not be reproduced. This was addressed through a number of measures, including much more stringent genome-wide significance thresholds [11], meta-analyses [12], pre-registration of study designs and protocols to reduce data dredging [13,14], and adoption of reporting standards such as STRENGTHENING the REporting of Genetic Association studies [15], all of which may be helpful in addressing AI-assisted exploitation of health data sources. Other measures employed in genomics, such as the use of large-scale datasets and the promotion of open data [16], may in contrast not work well in the AI-assisted era, as open-access naturally facilitates the type of data dredging and HARKING that has been seen in the last 3 years [17,18].

In this work, we attempt to establish the impact of AI-assisted formulaic templates we believe are currently in use, by examining trends in the number of publications published in journals and indexed in OpenAlex. We use these findings to discuss whether existing guidelines from policymakers are futureproofed to deal with these new strategies for the mass production of manuscripts, a crucial issue given the efforts to encourage Open Science and adoption of FAIR Guiding Principles [19–22]. In the worst-case scenario, unchecked exploitation and manipulation of FAIR assets could undermine confidence in publication-based dissemination and lead to researchers turning away from open data altogether.

2. Methods

To identify health data sources that might be current targets of exploitation, a search of the OpenAlex database via API using the Entrez module from the Biopython library was conducted (version 1.85) for publications dated between 2014 and 2025 [23], related to a list of 36 health or biomedical databases (Table 1) based on the authors' collective expertise in the field. Alternative search terms for each dataset were used where appropriate for acronyms, or additional Boolean operators were used where acronyms were associated with other issues (eg the PRoteomics IDentifications database of proteomics mass spectrometry information) and the search strings are detailed in Supplementary Materials, Table S1. An additional query was submitted to obtain combined publication counts for “at risk” datasets, using OR logic across dataset terms with deduplication. As a result, totals of combinations of datasets would be expected to be lower than the arithmetic sum of per-dataset counts. All searches were conducted for title or abstract, filtered to show type = article, and were run on January 3, 2026.

The period from 2014 to 2022 was taken as a baseline largely undisturbed by the use of large language models (LLMs) and other forms of AI-assisted workflows, consistent with our previous work identifying an acceleration from late-2022 onwards [4]. 2023–2025 was considered to be the period substantially affected by technology-driven productivity gains from paper mills. To quantify the deviation from trend, for instance, where a data source might be experiencing a natural growth trend for nonexploitative reasons, forecasts were constructed using Autoregressive Integrated Moving Average (ARIMA). ARIMA is a time series forecasting algorithm with three components: autoregression (using past values), differencing (to account for trends and seasonality), and moving averages (to remove noise). The excess production of manuscripts was then taken as the difference between the observed production of manuscripts and the ARIMA forecasts. The ARIMA model used parameters of autoregressive order $p = 1$, degree

of differencing $d = 1$, and moving average order $q = 1$. CIs were also constructed for the forecasts. ARIMA models were implemented in Python using the statsmodels library (version 0.14.4) [24].

Datasets where the publication rate in 2025 exceeded the 95% CI for the ARIMA forecast were identified as potentially being exploited by new technologies—and therefore worthy of further investigation. “Genomics” was used as a control term in the OpenAlex database, as a more mature field where extensive prior work has been performed by the research industry to reduce false discoveries and implausible conclusions. Secondary analyses were conducted to identify whether manuscript titles were becoming more homogenous/formulaic, and to identify datasets where there had been a geographic shift in the affiliations of last-named authors. For the identification of potentially formulaic titles, a simple count of increased frequency of words was conducted to test whether titles were becoming more homogenous. This was performed in Python using a count vectorizer function from the scikit-learn library (version 1.6.1) [25]. For the geographic analysis, simple count of geographic origin was used together with fold changes of countries of origin between 2022 and 2025.

3. Results

The fold changes in publications for each data source for the period between 2022 and 2025 are summarized in Table 2. Of the searched datasets, four were excluded for having insufficient publication counts to support later modeling steps (UK10K, World Bank Health Data, OpenSAFELY, and High-Resolution Intensive Care Unit Dataset) and the All of Us Research Programme was excluded as its Researcher Workbench was only made available for data access in late 2020 to US researchers and for international researchers in late 2023 [26]. Nine data sources (FinnGen, FDA Adverse Event Reporting System [FAERS], NHANES, UK Biobank, Global Burden of Disease, Centers for Disease Control and Prevention Wides-ranging Online Data for Epidemiologic Research [CDC WONDER], TriNetX, China Health and Retirement Longitudinal Study [CHARLS], and Medical Information Mart for Intensive Care [MIMIC] III/IV) met the criterion of exceeding the 95% CI for ARIMA forecasts. The median fold change was 1.2x, whereas the “genomics [title/abstract]” control search term produced a 2022–2025 fold change of 1.4x.

The deviations from the ARIMA-estimated trend are shown in Figure 1.

The data sources showing trend changes above the 95% CIs of forecasts were then analyzed for 2022 and 2025 to identify whether there was any change in titles of manuscripts (as a simple test for formulaic manuscript production). Nine datasets showed a sharp increase in certain tokens (words or phrases), as shown in Table 3, and this

Table 1. Data sources investigated, listed in alphabetic order

Dataset	Full name	API availability
1000 Genomes Project	1000 Genomes Project	Data can be accessed through various APIs and file transfer protocols
ABCD Study	Adolescent Brain Cognitive Development Study	Data access requires application; no public API available
ADNI	Alzheimer's Disease Neuroimaging Initiative	Data access requires application; no public API available
All of Us Research Program	All of Us Research Program	Data access requires application; no public API available
Biobank Japan	Biobank Japan Project	Data access requires application; no public API available
CCLE	Cancer Cell Line Encyclopedia	Data available for download; no dedicated API
CDC WONDER	Wide-ranging Online Data for Epidemiologic Research	Public-facing API using XML-based POST requests
CHARLS	China Health and Retirement Longitudinal Study	Data access requires application; no public API available
dbGaP	Database of Genotypes and Phenotypes	Only summary data and variable descriptions available as open access, individual data are controlled access. API available.
DHS	Demographic and Health Surveys	Data access requires registration; no public API available
eICU	eICU Collaborative Research Database	Data available for download; no dedicated API.
FAERS	Food and Drug Administration (FDA) Adverse Event Reporting System	Accessible via the openFDA API.
FinnGen	FinnGen Study	Data access requires application; no public API available, except for GWAS which is open access and has OpenGWAS APIs
Global Burden of Disease	Global Burden of Disease Study	Data available for download; no dedicated API
gnomAD	Genome Aggregation Database	Provides API access for querying genetic variant data
GTEx	Genotype-Tissue Expression Project	Data available for download; no dedicated API
HiRID	High-Resolution Intensive Care Unit Dataset	Data available for download; no dedicated API
HRS	Health and Retirement Study	Data access requires registration; no public API available
Human Connectome Project	Human Connectome Project	Data access requires application; no public API available
Human Microbiome Project	Human Microbiome Project	Data available for download; no dedicated API
ICGC	International Cancer Genome Consortium	Provides API access for genomic data queries
MetaboLights	MetaboLights	Offers RESTful API for accessing metabolomics data
Metabolomics Workbench	Metabolomics Workbench	Data available for download; no dedicated API
MIMIC III or IV	Medical Information Mart for Intensive Care	Data access requires credentialing; no public API available
NHANES	National Health and Nutrition Examination Survey	Data available for download via CDC website downloads, API endpoints, and R/Python packages
OpenSAFELY	OpenSAFELY Platform	Data access requires application; no public API available.
PhysioNet	PhysioNet (Research Resource for Complex Physiologic Signals)	Offers tools for data analysis but lacks a unified API.
PRIDE	PRoteomics IDentifications Database	Provides API access for proteomics data retrieval.
SEER	Surveillance, Epidemiology, and End Results Program	Data access requires application; no public API available.
SHARE	Survey of Health, Aging and Retirement in Europe	Data access requires registration; no public API available
TCGA	The Cancer Genome Atlas	Data accessible via the Genomic Data Commons API
TriNetX	TriNetX	Representational State Transfer API, only accessible to those with an institutional license (universities, pharma companies, or large hospitals)
UK Biobank	UK Biobank	Data access requires application; no public API available, except for GWAS which is open access and has

(Continued)

Table 1. Continued

Dataset	Full name	API availability
		OpenGWAS APIs
UK10K	UK10K Project	Data available for download; no dedicated API
WHO Global Health Observatory	World Health Organization Global Health Observatory	Offers API access to a wide range of global health data
World Bank Health Data	World Bank Health Data	Data available for download; no dedicated API

CDC, Centers for Disease Control and Prevention.

was especially marked for CDC WONDER (where 69% of titles commented on “trends” in 2025, compared with 20% of titles in 2022. Analysis of the control search term “genomics” showed that no tokens saw a change of more than 2.5% in frequency between 2022 and 2025, in contrast to the larger changes seen in the datasets suspected of exploitation. A list of the top 20 token changes is included in Supplementary Material, Table S3, including words which declined in usage.

The nine data sources were then examined for changes to geographic origin, focusing on country of affiliation for the last-named author on each publication. The growth in publications is shown as a choropleth in Figure 2. The largest change was for publications with affiliations located in China, which increased from 27% of publications indexed in the OpenAlex database in 2022 to 45% in 2025, or—in absolute terms—growth of 8679 publications. Over the same period, the share of publications originating in the United States declined from 25% to 19%, as its annual publication count increased by 2408 over the 3 years, with nearly half of this increase ($n = 1104$) in publications analyzing TriNetX.

Although the geographical split of growth in publications was very marked, it was not distributed evenly over the nine fastest growing datasets. Seven saw a sharp increase in publications from China, based on full attribution of affiliation (Table 4). This pattern was particularly evident for FAERS, where 66% of publications had an affiliation from China in 2025. Pakistan also showed a large change relative to its previous number, but this was focused largely within the CDC WONDER dataset. The rapid growth in publications using TriNetX was concentrated in the United States.

4. Discussion

Many approaches have been proposed for the identification of paper mill outputs or other problematic research practices, either reviewing individual papers [17] or seeking to identify concerning issues in overall publication trends (such as evidence of manipulated citation networks or retractions) [27,28]. Here we use a scientometric approach to demonstrate that there has been rapid growth in numbers of publications since 2022 analyzing open-source datasets, with nine datasets seeing both a break with previous trends that exceeded a 95% forecast CI and signs of formulaic or

templated research. This break in trend may also have been accelerated by the development of GenAI in late 2022 [29], but it is notable that the acceleration in publication rates differs between data sources. NHANES, FinnGen, and the Global Burden of Disease Study experienced a break with trend in 2023, but FAERS publications saw an acceleration vs. the previous trend only in 2024, with TriNetX and CDC WONDER seeing a larger acceleration in 2025. This may be suggestive of datasets being discovered at different times, albeit there are no obvious signs that datasets are being abandoned as other datasets start to be exploited.

Although comprehensive analysis of individual publications was beyond the scope of this work, many showed signs of titles becoming more formulaic, with “fingerprint” methodologies, words and phrases consistent with previously described trends in the production of low-quality manuscripts. Across the datasets, there was a strong trend toward reporting the name of the dataset used, and either “association”, “trends”, or “mortality”. This is consistent with our previous work which showed publications taking a well-described condition such as type 2 diabetes mellitus, which is multifactorial in nature, and then analyzing a single indicator or ratio for simple analysis and reporting the trend or association. Such simplistic studies fail to capture the complex and multifactorial nature of exposome-phenome associations [30]. The increased frequency of specific tokens and methodologies (eg the majority of the studies investigating CDC WONDER used Joinpoint software and the same workflow) is suggestive of a template-driven improvement in productivity and output, possibly also using GenAI [31].

Coincident with these changes, there was also a dramatic shift in geographic origin of corresponding (last-named) author affiliations for the 9 most affected databases. The absolute increase for China between 2022 and 2025 (8679 publications) dwarfed that for the United States (2408), with the next three largest increases from Pakistan (708), the United Kingdom (383), and India (264). Authors with affiliations to Chinese hospitals have been associated with high retraction rates of papers in various fields [32], and researchers from China are also subject to different incentives [33], as are scientists in lower-middle income countries [34], often due to difficulties in undertaking and publishing research and a lack of institutional support. Furthermore, the Chinese supreme court has recently issued guidance on paper mills and on scientific fraud, calling for lower

Table 2. Fold changes in publication numbers between 2022 and 2025 according to health data source, split between those with a change > median vs. those with a change < median and ranked largest to smallest

Data sources with fold change > median (median = 1.2)	Fold change	Data sources with fold change ≤ median (median = 1.2)	Fold change
CDC WONDER	17.4	1000 Genomes Project	1.2
TriNetX	8.0	BioBank Japan	1.1
All of Us Research Program	6.9	OpenSAFELY	1.1
FAERS	5.6	DHS	1.1
FinnGen	5.5	gnomAD	1.1
Global Burden of Disease	3.2	SEER	1.1
CHARLS	2.9	GTE _x	1.0
HiRID	2.5	SHARE	1.0
NHANES	2.4	Human Connectome Project	1.0
UK Biobank	2.3	HRS	1.0
MIMIC-III or MIMIC-IV	2.3	PRIDE	0.9
ABCD Study	2.1	TCGA	0.8
WHO Global Health Observatory	2.0	PhysioNet	0.7
dbGaP	1.9	CCLC	0.6
MetaboLights	1.9	Human Microbiome Project	0.6
eICU Collaborative Research Database	1.7	ICGC	0.5
Metabolomics Workbench	1.4	UK10K	0.4
ADNI	1.3	World Bank Health Data	0.0

MIMIC, Medical Information Mart for Intensive Care; CHARLS, China Health and Retirement Longitudinal Study; NHANES, National Health And Nutrition Examination Survey; CDC WONDER, Centers for Disease Control and Prevention; FAERS, FDA Adverse Event Reporting System; HiRID, High-Resolution Intensive Care Unit Dataset; ABCD, Adolescent Brain Cognitive Development; WHO, World Health Organization; dbGaP, Database of Genotypes and Phenotypes; ADNI, Alzheimer's Disease Neuroimaging Initiative; DHS, Demographic and Health Surveys; gnomAD, Genome Aggregation Database; SEER, Surveillance, Epidemiology, and End Results Program; GTE_x, Genotype-Tissue Expression Project; SHARE, Survey of Health, Aging and Retirement in Europe; HRS, Health and Retirement Study; PRIDE, PRoteomics IDentifications; TCGA, The Cancer Genome Atlas; CCLC, Cancer Cell Line Encyclopedia; ICGC, International Cancer Genome Consortium.

courts to crack down on the mass-produced academic paper industry [35].

Overall, of the datasets showing an acceleration in publication rates, the additional evidence of formulaic titles and a geographic shift in authorship is strongly suggestive of a change in publication patterns for nine of the datasets analyzed in this work. These are FAERS, NHANES, FinnGen, UK Biobank, the Global Burden of Disease Study, CHARLS, MIMIC III/IV, TriNetX, and CDC WONDER. The rapid changes over the 4-year period may well be due to paper mill exploitation. Eight of the nine datasets also share open access elements, often with APIs that facilitate highly productive workflows. For the UK Biobank and FinnGen, open-access GWAS data were typically used [36], NHANES has a well-developed API system [37], FAERS research is facilitated through OpenVigil [38], and although the Global Burden of Disease Study does not have an API, the data are freely available for download. We have written previously on two-sample Mendelian randomization and association templates, but drug safety is a newly exploited type of data. FAERS itself is a valuable asset in a field that has not always had full data transparency [39], but is a voluntary reporting system and so cannot provide estimates

of incidence, cannot analyze causality, and additionally has considerable potential for bias based on physician or public preferences [40]. Misleading publications based on such drug safety reports pose a particular risk to the public, either by amplifying unnecessary concerns or underplaying risks. TriNetX presents a different profile, with many of the publications targeted at conferences and a US-centric production model focused on medical schools. This illustrates that the demand for mass-produced manuscripts is not an issue related solely to the Global South.

Such problematic outputs present clear harms. Taken individually, the papers we find are simply not very interesting (they are formulaic and repeat the same analyses across thousands of variables), but taken together they become misleading through the introduction of false discoveries, dilution of genuine findings in the scientific literature, and reduced credibility for high quality data sources. An additional issue is that many GenAI models are trained on—and learning from—scientific literature. With the inputs for learning potentially becoming corrupted, these models will propagate false science. For the nine most problematic datasets identified here, the excess publications above trend (around twelve thousand) represented 50% of

Forecast vs Actual Trends (Selected Datasets)

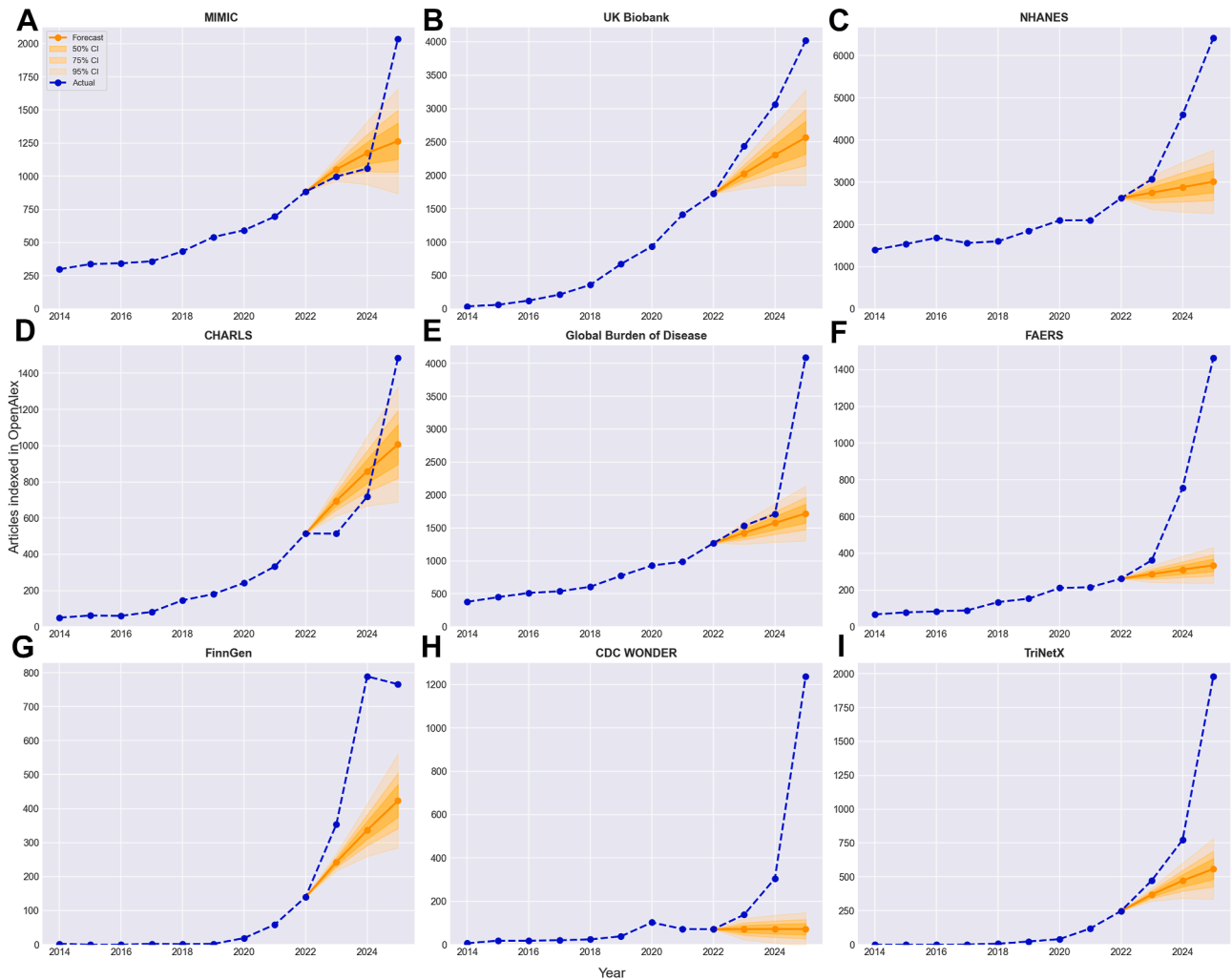


Figure 1. Actual publication counts compared with ARIMA forecasts, using 2014–2022 as the training period and 2023–2025 as the forecast period (A) MIMIC III/IV (B) UK Biobank (C) NHANES (D) CHARLS (E) Global Burden of Disease Study (F) FAERS (G) FinnGen (H) CDC WONDER, and (I) TriNetX. Numerical data underlying these figures are included in Supplementary Materials, [Table S2](#). ARIMA, AutoRegressive Integrated Moving Average; MIMIC, Medical Information Mart for Intensive Care; CHARLS, China Health and Retirement Longitudinal Study; NHANES, National Health And Nutrition Examination Survey; CDC WONDER, Centers for Disease Control and Prevention; FAERS, FDA Adverse Event Reporting System.

the 2025 output, a significant proportion to be potentially misleading.

We and others have previously written on measures to mitigate against formulaic manuscript production, focusing on more frequent use of desk rejections by journals to reduce burdens on peer reviewers, dedicated statistical reviewers [41], use of application numbers by data providers, and more effective postpublication correction [42,43]. Others have also written on the need for effective tools to identify problematic manuscripts such as the recently proposed GRABDROP checklist [44], and we suggest that increased awareness of datasets currently experiencing worrying trends and key phrases associated with formulaic outputs will assist in this process [7,27,45]. It is, however, important to stress that the threat to scientific integrity identified here presents risks to the wider goals of Open Science

and compliance with the FAIR Guiding Principles [19,46]. The factors that drive compliance with these principles also make such datasets vulnerable to exploitation. To preserve the overall goal of Open Science and FAIR compliance, we believe that unrestricted open access to AI-ready data may not be the best option for all data resources. For some, approved access and/or pre-registration may offer safeguards against exploitative data dredging. A more radical solution given the ease of analyzing simple relationships such as “The association of [Predictor A] with [Outcome B] in [Population C] using a cross-sectional national dataset [Open Access Dataset D]” would be for the scientific community to deem such results as effectively already available online, and so not suitable for publication.

A number of limitations should be stressed in this work, the most significant of which is that this is a scientometric

Table 3. Increased incidence of tokens, measured by change in % of titles where each token was used

Dataset	Title tokens	2022 % of titles featuring token	2025 % of titles featuring token	Change %
CDC Wonder	trends	19.7	68.8	49.1
	cdc wonder	7.0	52.7	45.6
	analysis	18.3	54.3	36.0
	mortality	52.1	88.0	35.9
	database	5.6	27.3	21.7
CHARLS	charls	9.1	29.3	20.1
	association	15.7	27.8	12.1
	index	2.3	14.3	12.0
	risk	11.1	22.5	11.5
	disease	4.5	13.1	8.6
FAERS	faers	26.0	54.8	28.9
	analysis	31.7	50.4	18.7
	database	20.2	37.5	17.2
	world	13.7	27.5	13.7
	safety	7.3	20.2	12.9
FinnGen	causal	17.1	30.7	13.5
	insights	0.0	7.8	7.8
	analysis	13.6	20.6	7.1
	potential	0.7	7.7	7.0
	relationship	2.9	9.7	6.8
Global Burden of Disease	analysis	20.6	41.8	21.2
	global burden of disease	25.1	44.7	19.6
	study	31.9	46.5	14.6
	regional	8.1	22.4	14.3
	trends	13.3	27.4	14.1
MIMIC III/IV	mimic-iv	7.9	29.1	21.2
	database	9.7	24.2	14.4
	retrospective	9.4	22.8	13.4
	mortality	18.3	31.1	12.7
	study	12.4	24.3	11.8
NHANES	nhanes	21.5	43.4	21.9
	association	26.2	41.4	15.2
	study	14.0	26.6	12.6
	index	5.5	16.9	11.4
	cross-sectional	8.2	18.5	10.3
TriNetX	trinetx	4.0	22.1	18.1
	study	18.1	32.4	14.3
	cohort	10.5	23.6	13.2
	matched	4.4	16.5	12.1
	propensity	4.8	15.7	10.8
UK Biobank	risk	28.5	32.5	4.0
	plasma	0.3	3.8	3.5
	insights	1.2	4.4	3.2
	cohort	14.9	17.9	3.0
	prospective	9.1	12.1	3.0

MIMIC, Medical Information Mart for Intensive Care; CHARLS, China Health and Retirement Longitudinal Study; NHANES, National Health And Nutrition Examination Survey; CDC WONDER, Centers for Disease Control and Prevention; FAERS, FDA Adverse Event Reporting System.

analysis, rather than a comprehensive review of individual papers. For instance, GenAI has the potential to generate formulaic low-quality outputs, sometimes referred to in other fields as “AI slop” [47]. It is challenging from the outside to tell the difference between deliberate, coordinated mass production by paper mills and uncoordinated mass production by individuals facilitated by LLMs, and this is a limitation of this research. A second limitation is that accelerations in publication may be due to good reasons. For instance, the UK Biobank has made a number of new datasets available over time. Third, this analysis is retrospective, and future behavior may change in response to recent publisher policy changes to restrict submissions dealing with open access datasets, such as NHANES [48,49]. Given the adaptive and adversarial nature of unethical actors, we expect both scientometric analyses and targeted investigations to form part of an ongoing effort to protect research integrity and the principles of Open Science, especially given that paper mills will adapt their strategies as existing approaches are brought to light [50,51]. It

should also be noted that in this work we have not integrated resources such as the Problematic Paper Screener; this may be an opportunity for future work, albeit these traditional flags (eg for the detection of tortured phrases) [52] are not typically found in this type of formulaic manuscript, possibly due to adaptive changes by paper mills. Finally, it should be noted that scientometric analysis does not substitute for existing methods of detecting problematic manuscripts, such as citation networks, evidence of image manipulation or text similarities, as well as the backstop of peer review and editorial assessment [17,53–55]. Nevertheless, the fact that this work reviews accepted manuscripts provides evidence that existing tools used by publishers may be inadequate for new challenges posed by paper mill [18,56,57], especially when postpublication corrections/retractions are often slow, or may not happen at all [58]. Furthermore, even retracted articles can contaminate evidence syntheses, and the sheer weight of publications flagged in this work is suggestive that these negative downstream impacts are likely to increase [59,60].

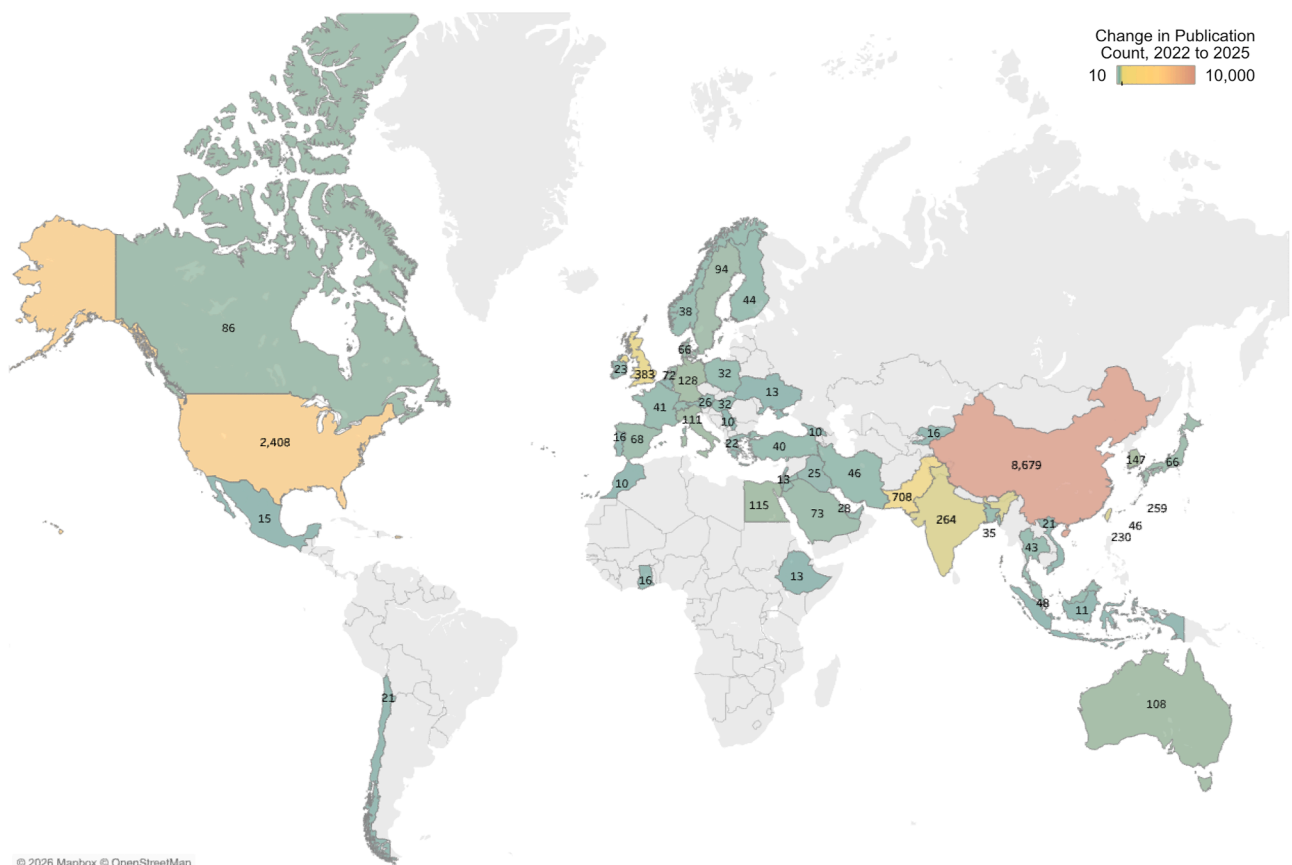


Figure 2. Choropleth (geographical heatmap) of 2022–2025 change in annual publication count by country of author affiliations (fully counted), for the following nine datasets: CDC WONDER, CHARLS, FAERS, NHANES, UK Biobank, FinnGen, Global Burden of Disease Study, MIMIC III/IV, TriNetX. Filtered to only show countries with an increase in publications > 10 indexed in OpenAlex. Color range is not linear and is centered at 200 publications. Data underlying this choropleth are shown in Supplementary Materials, Table S4. As author affiliations are fully counted rather than fractionally, the total of affiliations does not sum to the number of publications. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.) MIMIC, Medical Information Mart for Intensive Care; CHARLS, China Health and Retirement Longitudinal Study; NHANES, National Health And Nutrition Examination Survey; CDC WONDER, Centers for Disease Control and Prevention; FAERS, FDA Adverse Event Reporting System.

Table 4. Table of fold changes for last-named author affiliations for the country seeing the largest absolute growth

Dataset	Country seeing largest fold change	2022 affiliations from largest fold change country	2025 affiliations from largest fold change country	Fold change
CDC WONDER	Pakistan	1	553	553.0
CHARLS	China	442	1208	2.7
FAERS	China	102	843	8.3
FinnGen	China	76	569	7.5
Global Burden of Disease Study	China	427	2389	5.6
MIMIC III/IV	China	381	1017	2.7
NHANES	China	856	4043	4.7
TriNetX	United States	205	1309	6.4
UK Biobank	China	463	1596	3.4

MIMIC, Medical Information Mart for Intensive Care; CHARLS, China Health and Retirement Longitudinal Study; NHANES, National Health And Nutrition Examination Survey; CDC WONDER, Centers for Disease Control and Prevention; FAERS, FDA Adverse Event Reporting System.

In conclusion, our scientometric analysis highlights significant concerns about research integrity, particularly in light of previously documented associations between institutional pressures and increased retraction rates and the risk of paper mills employing new AI-supported workflows to produce manuscripts on an industrial scale. This growth poses a direct threat to the core principles of scientific rigor, and those of Open Science and FAIR data practices, given that the accessibility and interoperability designed to facilitate legitimate research also enables exploitation. To mitigate these risks and safeguard scientific integrity, we advocate for controlled data-access models paired with mandatory pre-registration of analyses, rather than unrestricted open access. This balanced approach is essential to preserving the intended benefits of Open Science while preventing its misuse, ensuring that scientific advancement continues to be reliable, reproducible, and trustworthy.

CRediT authorship contribution statement

Matt Spick: Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anthony Onoja:** Software, Data curation. **Charlie Harrison:** Methodology, Investigation, Data curation. **Stefan Stender:** Writing – review & editing, Methodology. **Jennifer Byrne:** Writing – review & editing, Methodology. **Nophar Geifman:** Writing – review & editing, Resources, Project administration, Conceptualization.

Declaration of competing interest

M.S. reports financial support was provided by UK Research and Innovation. C.H. reports financial support

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112203>.

Data availability

All data used in the preparation of this manuscript are included in Supplementary Materials. All code used in this work employed standard Python libraries without any modifications. The underlying data are also maintained and updated at the following Tableau Public dashboard <https://public.tableau.com/app/profile/matt.spick/viz/TheExploitationofOpenScience/OpenScienceExploitationDashboard>

References

[1] Christopher J. The raw truth about paper mills. *FEBS Lett* 2021;595: 1751–7. <https://doi.org/10.1002/1873-3468.14143>.
[2] Netzer NC. Artificial intelligence – the Janus-faced tool in our hands. *Sleep Breath* 2024;28:1861–2. <https://doi.org/10.1007/s11325-024-03129-7>.
[3] Richardson RAK, Hong SS, Byrne JA, Stoeger T, Amaral LAN. The entities enabling scientific fraud at scale are large, resilient, and growing rapidly. *Proce Nat Acad Sci* 2025;122:e2420092122. <https://doi.org/10.1073/pnas.2420092122>.

- [4] Suchak T, Aliu AE, Harrison C, Zwiggelaar R, Geifman N, Spick M. Explosion of formulaic research articles, including inappropriate study designs and false discoveries, based on the NHANES US National Health Database. *PLoS Biol* 2025;23:e3003152. <https://doi.org/10.1371/journal.pbio.3003152>.
- [5] Byrne JA, Stender S. More science friction for less science fiction. *PLOS Biol* 2025;23:e3003167. <https://doi.org/10.1371/journal.pbio.3003167>.
- [6] Grimes DR, Heathers J. “Cake causes herpes?” - promiscuous dichotomisation induces false positives. *BMC Med Res Methodol* 2025;25:255. <https://doi.org/10.1186/s12874-025-02712-0>.
- [7] Byrne JA, Abalkina A, Akinduro-Aje O, Christopher J, Eaton SE, Joshi N, et al. A call for research to address the threat of paper mills. *PLoS Biol* 2024;22:e3002931. <https://doi.org/10.1371/journal.pbio.3002931>.
- [8] Mainous AG. Papermills as another challenge to research integrity and trust in science. *Front Med* 2025;12:1557024. <https://doi.org/10.3389/fmed.2025.1557024>.
- [9] Stender S, Gellert-Kristensen H, Smith GD. Reclaiming mendelian randomization from the deluge of papers and misleading findings. *Lipids Health Dis* 2024;23:286. <https://doi.org/10.1186/s12944-024-02284-w>.
- [10] Smith GD, Ebrahim S. Mendelian randomisation at 20 years: how can it avoid hubris, while achieving more? *Lancet Diabetes Endocrinol* 2024;12:14–7. [https://doi.org/10.1016/S2213-8587\(23\)00348-0](https://doi.org/10.1016/S2213-8587(23)00348-0).
- [11] Ioannidis JPA, Thomas G, Daly MJ. Validating, augmenting and refining genome-wide association signals. *Nat Rev Genet* 2009;10:318–29. <https://doi.org/10.1038/nrg2544>.
- [12] Evangelou E, Ioannidis JPA. Meta-analysis methods for genome-wide association studies and beyond. *Nat Rev Genet* 2013;14:379–89. <https://doi.org/10.1038/nrg3472>.
- [13] Nosek BA, Ebersole CR, DeHaven AC, Mellor DT. The preregistration revolution. *Proc Natl Acad Sci* 2018;115:2600–6. <https://doi.org/10.1073/pnas.1708274114>.
- [14] Chambers CD, Dienes Z, McIntosh RD, Rotshtein P, Willmes K. *Registered reports*: realigning incentives in scientific publishing. *Cortex* 2015;66:A1–2. <https://doi.org/10.1016/j.cortex.2015.03.022>.
- [15] Little J, Higgins JPT, Ioannidis JPA, Moher D, Gagnon F, von Elm E, et al. STrengthening the REporting of genetic Association studies (STREGA)— an extension of the STROBE statement. *PLoS Med* 2009;6:e1000022. <https://doi.org/10.1371/journal.pmed.1000022>.
- [16] Manolio TA, Fowler DM, Starita LM, Haendel MA, MacArthur DG, Biesecker LG, et al. Bedside back to bench: building bridges between basic and clinical genomic research. *Cell* 2017;169:6–12. <https://doi.org/10.1016/j.cell.2017.03.005>.
- [17] Abalkina A. Publication and collaboration anomalies in academic papers originating from a paper mill: evidence from a Russia-based paper mill. *Learned Publishing* 2023;36:689–702. <https://doi.org/10.1002/leap.1574>.
- [18] Liverpool L. AI intensifies fight against “paper mills” that churn out fake research. *Nature* 2023;618:222–3. <https://doi.org/10.1038/d41586-023-01780-w>.
- [19] Wilkinson MD, Dumontier M, Aalbersberg IJ, Appleton G, Axton M, Baak A, et al. The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* 2016;3:160018. <https://doi.org/10.1038/sdata.2016.18>.
- [20] Foster ED, Deardorff A. Open science framework (OSF). *J Med Libr Assoc* 2017;105:203–6. <https://doi.org/10.5195/jmla.2017.88>.
- [21] McKiernan EC, Bourne PE, Brown CT, Buck S, Kenall A, Lin J, et al. How open science helps researchers succeed. *eLife* 2016;5:e16800. <https://doi.org/10.7554/eLife.16800>.
- [22] Mirowski P. The future(s) of open science. *Soc Stud Sci* 2018;48:171–203. <https://doi.org/10.1177/0306312718772086>.
- [23] Biopython Biopython n.d. Available at: <https://biopython.org/>. Accessed March 26, 2025.
- [24] Seabold S, Perktold J. “Statsmodels: Econometric and statistical modeling with python.” *Proceedings of the 9th Python in Science Conference*. 2010. Available at: <https://www.statsmodels.org/stable/index.html#citation>. Accessed March 12, 2026.
- [25] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in python. *J Machine Learn Res* 2011;12:2825–30.
- [26] The All of Us Research Program Investigators. The “All of Us” Research Program. *N Engl J Med* 2019;381:668–76. <https://doi.org/10.1056/NEJMs1809937>.
- [27] Candal-Pedreira C, Ross JS, Ruano-Ravina A, Egilman DS, Fernández E, Pérez-Ríos M. Retracted papers originating from paper mills: cross sectional study. *BMJ* 2022;379:e071517. <https://doi.org/10.1136/bmj-2022-071517>.
- [28] Candal-Pedreira C, Guerra-Tort C, Ruano-Ravina A, Freijedo-Farinas F, Rey-Brandariz J, Ross JS, et al. Retracted papers originating from paper mills: a cross-sectional analysis of references and citations. *J Clin Epidemiol* 2024;172:111397. <https://doi.org/10.1016/j.jclinepi.2024.111397>.
- [29] Liu Y, Han T, Ma S, Zhang J, Yang Y, Tian J, et al. Summary of ChatGPT-Related research and perspective towards the future of large language models. *Meta-Radiology* 2023;1:100017. <https://doi.org/10.1016/j.metrad.2023.100017>.
- [30] Patel CJ, Ioannidis JP, Manrai AK. The architecture of exposome-phenome associations. *medRxiv* 2025. <https://doi.org/10.1101/2025.06.05.25329055>.
- [31] Maupin D, Suchak T, Barnett A, Spick M. Dramatic increases in redundant publications in the Generative AI era. *BMC Med* 2025;24:29. <https://doi.org/10.1186/s12916-025-04569-y>.
- [32] Van Noorden R. Exclusive: these universities have the most retracted scientific articles. *Nature* 2025;638:596–9. <https://doi.org/10.1038/d41586-025-00455-y>.
- [33] Mallapaty S. China conducts first nationwide review of retractions and research misconduct. *Nature* 2024;626:700–1. <https://doi.org/10.1038/d41586-024-00397-x>.
- [34] Vasconez-Gonzalez J, Izquierdo-Condoy JS, Naranjo-Lara P, Garcia-Bereguini MÁ, Ortiz-Prado E. Integrity at stake: confronting “publish or perish” in the developing world and emerging economies. *Front Med (Lausanne)* 2024;11:1405424. <https://doi.org/10.3389/fmed.2024.1405424>.
- [35] Mallapaty S. China’s supreme court calls for crack down on paper mills. *Nature* 2025;639:285–6. <https://doi.org/10.1038/d41586-025-00612-3>.
- [36] Hartwig FP, Davies NM, Hemani G, Davey Smith G. Two-sample Mendelian randomization: avoiding the downsides of a powerful, widely applicable but potentially fallible technique. *Int J Epidemiol* 2016;45:1717–26. <https://doi.org/10.1093/ije/dyx028>.
- [37] Ale L, Gentleman R, Sonmez TF, Sarkar D, Endres C. nhanesA: achieving transparency and reproducibility in NHANES research. *Database* 2024;2024:baae028. <https://doi.org/10.1093/database/baae028>.
- [38] OpenVigil pharmacovigilance search engines n.d. Available at: <https://openvigil.sourceforge.net/>. Accessed May 2, 2025.
- [39] Golder S, Loke YK. Is there evidence for biased reporting of published adverse effects data in pharmaceutical industry-funded studies? *Br J Clin Pharmacol* 2008;66:767–73. <https://doi.org/10.1111/j.1365-2125.2008.03272.x>.
- [40] Chedid V, Vijayvargiya P, Camilleri M. Advantages and limitations of the federal adverse events reporting system in assessing adverse event reporting for eluxadoline. *Clin Gastroenterol Hepatol* 2018;16:336–8. <https://doi.org/10.1016/j.cgh.2017.11.025>.
- [41] Spick M, Higgins J, Green CL, Matsouaka R, Shin DB, Hall RP, et al. Observations from statistical review editors: a commentary. *JID Innov* 2024;4:100302. <https://doi.org/10.1016/j.xjidi.2024.100302>.
- [42] Ortega JL. Classification and analysis of PubPeer comments: how a web journal club is used. *J Assoc Inf Sci Technology* 2022;73:655–70. <https://doi.org/10.1002/asi.24568>.
- [43] Serrano-Puche J, Rodríguez-Salcedo N, Martínez-Costa M-P. Trust, disinformation, and digital media: Perceptions and expectations about

- news in a polarized environment. Profesional de La Información 2023; 32. <https://doi.org/10.3145/epi.2023.sep.18>.
- [44] Rudan I, Song P, Adeyoye D, Campbell H. Journal of Global Health's Guidelines for Reporting Analyses of Big Data Repositories Open to the Public (GRABDROP): preventing "paper mills", duplicate publications, misuse of statistical inference, and inappropriate use of artificial intelligence. J Glob Health 2025;15:01004. <https://doi.org/10.7189/jogh.15.01004>.
- [45] Parker L, Boughton S, Bero L, Byrne JA. Paper mill challenges: past, present, and future. J Clin Epidemiol 2024;176:111549. <https://doi.org/10.1016/j.jclinepi.2024.111549>.
- [46] Jacobsen A, De Miranda Azevedo R, Juty N, Batista D, Coles S, Cornet R, et al. FAIR principles: interpretations and implementation considerations. Data Intell 2020;2:10–29. https://doi.org/10.1162/dint_r_00024.
- [47] Martin A, Newell B. Synthetic data, synthetic media, and surveillance. S S 2024;22. <https://doi.org/10.24908/ss.v22i4.18334>.
- [48] Journals and publishers crack down on research from open health data sets n.d. Available at: <https://www.science.org/content/article/journals-and-publishers-crack-down-research-open-health-data-sets>. Accessed October 27, 2025.
- [49] Taylor L. AI: journals are automatically rejecting public health dataset papers to combat paper mills. BMJ 2025;391:r2170. <https://doi.org/10.1136/bmj.r2170>.
- [50] Maupin D, Spick M, Geifman N. Safeguarding open science from exploitative practices. PLoS Med 2025;22:e1004851. <https://doi.org/10.1371/journal.pmed.1004851>.
- [51] Lumbard H, Routledge D. Open science and transparency are our strongest tools in the fight against fraudulent publishing activities. PLoS Med 2025;22:e1004774. <https://doi.org/10.1371/journal.pmed.1004774>.
- [52] Cabanac G, Labbé C, Magazinov A. Tortured phrases: a dubious writing style emerging in science. Evidence of critical issues affecting established journals. arXivOrg. 2021. Available at: <https://arxiv.org/abs/2107.06751v1>. Accessed October 27, 2025.
- [53] van Diest RA, Seifert R, van der Heyden MAG. An extra pair of eyes: adopting innovative approaches to detect integrity issues in Naunyn-Schmiedeberg's archives of pharmacology. Naunyn-Schmiedeberg's Arch Pharmacol 2025;398:1–8. <https://doi.org/10.1007/s00210-024-03697-1>.
- [54] Sanderson K. Science's fake-paper problem: high-profile effort will tackle paper mills. Nature 2024;626:17–8. <https://doi.org/10.1038/d41586-024-00159-9>.
- [55] Byrne JA, Abalkina A, Christopher J, Soulière MF. Rethinking peer review using the Swiss cheese model to better flag problematic manuscripts. Learned Publishing; 2025;38. <https://doi.org/10.1002/leap.2021>.
- [56] Maupin D, Suchak T, Barnett A, Spick M. Dramatic increases in redundant publications in the generative AI era. BMC Med 2025; 24:29. <https://doi.org/10.1101/2025.09.09.25335401>.
- [57] Abalkina A, Aquarius R, Bik E, Bimler D, Bishop D, Byrne J, et al. "Stamp out paper mills" — science sleuths on how to fight fake research. Nature 2025;637:1047–50. <https://doi.org/10.1038/d41586-025-00212-1>.
- [58] Naddaf M. Journal targeted by paper mill still grappling with the aftermath years later. Nature 2025. <https://doi.org/10.1038/d41586-025-01010-5>.
- [59] Tang G, Cai H. Citation contamination by paper mill articles in systematic reviews of the life sciences. JAMA Netw Open 2025;8:e2515160. <https://doi.org/10.1001/jamanetworkopen.2025.15160>.
- [60] Gross CP, Flanagan A, Perencevich EN, Inouye SK. Mitigating the impact of retracted studies in the medical literature. JAMA Intern Med 2025;185:621–2. <https://doi.org/10.1001/jamainternmed.2025.0251>.